superconducting quantum hardware. These concepts provide the foundation for everything that follows.

However, we have only just begun our journey, and many important questions remain unanswered. Why are quantum computers powerful? Why is a single qubit not enough? Why does quantum phase matter? And what makes quantum algorithms fundamentally different from classical probabilistic algorithms? Don't worry—we have just started flexing our muscles. In the coming articles, we will answer each of these questions.

The defining feature of this series is its three-level approach to understanding quantum computing—examining every important concept from the perspectives of programming, mathematics, and hardware. This integrated approach is rarely found in introductory tutorials or textbooks, which often focus on only one of these aspects. Another promise of this series is to explore not only the technical foundations of quantum computing but also its history, current developments, future directions, and the many misconceptions that surround this fascinating field. So, before we move any further, let us begin by bursting two of the most common myths about quantum computing—myths that are widely repeated but fundamentally incorrect.

One of the most common myths about quantum computing is that a quantum computer can "try every possible answer simultaneously." Although this sounds fascinating, it is not an accurate description of how quantum computers work. A qubit can indeed exist in a superposition of states, allowing a quantum system to represent many possibilities at once. However, when the qubits are measured, only one outcome is obtained—not all possible answers. If quantum computers could simply examine every possible solution and read out the correct one, many computationally difficult problems would become trivial, which is certainly not the case.

The real power of quantum computing lies not in trying every answer simultaneously, but in manipulating quantum states so that the correct answer becomes more likely to appear when the final measurement is made. How this happens is one of the central ideas in quantum computing, involving concepts such as quantum phase, interference, and multi-qubit systems.

The second myth is that quantum computers will replace classical computers. Quantum computers are not designed to replace the laptops, desktops, smartphones, or servers that we use every day. In fact, everyday tasks such as web browsing, word processing, video streaming, email, and gaming are performed far more efficiently by classical computers. For such applications, quantum computers offer little or no advantage.

Instead, quantum computers are expected to serve as highly specialised machines that complement rather than replace classical computers, much like GPUs today accelerate graphics processing and artificial intelligence. They are likely to excel at specific classes of problems, including quantum simulation, cryptography, optimization, and certain machine learning tasks, while conventional computers continue to handle most of the everyday computing. The future of computing is therefore unlikely to be a choice between classical and quantum computing. Rather, it will be a partnership in which classical and quantum computers work together, each solving the problems for which it is best suited.

Now, let us turn our attention to one of the most fundamental—and often one of the most challenging—concepts in quantum computing: quantum phase. To develop an intuitive understanding of this idea, we shall study the two remaining Pauli gates: Pauli-Y and Pauli-Z. Let us begin with the lesser of these two evils—the Pauli-Z gate. I jokingly refer to them as 'evil' because they are notoriously difficult to understand and typically require much more explanation than the Pauli-X gate. Fortunately, there is an excellent visual aid that makes these concepts far more intuitive. As in the previous two articles, we shall once again make use of the online Bloch sphere simulator available at *https://bloch.kherb.io/*. By observing how these gates rotate the state vector on the Bloch sphere, the role of quantum phase becomes much easier to appreciate.

Open the Bloch sphere simulator and, from the Quantum gates menu, click the Z button corresponding to the Pauli-Z gate. What did you observe? At first glance, it appears that nothing happened. The arrow still points towards the north pole of the Bloch sphere, just as it did before. Is something wrong with the simulator? Not at all. In fact, the simulator is behaving exactly as it should.

The Pauli-Z gate performs a 180° (π radians) rotation about the z-axis of the Bloch sphere. Since the qubit is initially in the state $|0\rangle$, its state vector already lies along the z-axis. Rotating an arrow about its own axis does not change its visible direction—just as spinning a perfectly straight pencil about its own length leaves it looking exactly the same. The rotation has indeed taken place, but it cannot be seen from the orientation of the arrow alone.

This hidden change is known as a phase change. Unlike a bit flip, which changes the observable state of a qubit, a phase change does not immediately alter what we see or what we obtain when we measure the qubit. Instead, it changes the relative phase of the quantum state. Although this phase is invisible by itself, it profoundly influences how the qubit behaves when subsequent quantum gates are applied. In other words, phase cannot be observed directly, but its effects become evident through later quantum operations. As we shall soon discover, this seemingly invisible property lies at